

Erin Atkinson

MP605
60 St. George Street
ON M5S 1A7
Toronto
Canada

✉ erin.atkinson@mail.utoronto.ca
erin-atkinson.github.io

Education

- 2022– **PhD Candidate, Physics**, *University of Toronto, Canada*,
Earth, Atmospheric and Planetary physics
Submesoscale ocean fronts, filaments and vortices in multiscale flows (tentative).
- 2021–2022 **MSci Natural Sciences**, *Fitzwilliam College, University of Cambridge, UK*,
Specialising in Experimental and Theoretical Physics
First Class Honours
Using unsupervised classification of ocean data to track changes in Southern Hemisphere winds.
- 2018–2021 **BA Natural Sciences**, *Fitzwilliam College, University of Cambridge, UK*,
Specialising in Experimental and Theoretical Physics
First Class Honours
- 2011–2018 *Macmillan Academy, UK*
○ A Level - 4 A* Physics, Chemistry, Mathematics, Further Mathematics
○ GCSE - 10 A*-B Including English Language and English Literature.

Publications

Erin Atkinson and Nicolas Grisouard. “Formation and Arrest of a Surface Density Front via Strain-Driven Frontogenesis”. In: *Geophysical Research Letters* 53.12 (2026), e2025GL119220

Erin Atkinson, James Cyrus McWilliams, and Nicolas Grisouard. “Near-inertial echoes of ageostrophic instability in submesoscale filaments”. In: *Journal of Fluid Mechanics* 1015 (2025), A17

Emma JD Boland, Erin Atkinson, and Dani C Jones. “A novel heuristic method for detecting overfit in unsupervised classification of climate model data”. In: *Environmental Data Science* 2 (2023), e46

Employment and volunteer positions

- 2022– **Various teaching assistant appointments**
Teaching undergraduate students enrolled in physics courses offered by the Department of Physics at the University of Toronto. Primarily leading practicals and tutorials in addition to marking exam scripts and writing official solutions.
- 2024–2025 **Graduate Liason Committee**
Representing the interests of physics graduate students in the department in academic, supervisory and social contexts.
- 2022–2023 **Doors Open Toronto, Science Rendezvous**
Regular presentation at outreach events.
- 2022–2023 **PGSA Internal Vice President**
Executive position in the Physics Graduate Student Association at the department. Primary responsibilities include:
- Liasing with external providers to organise the daily coffee hour
 - Organising social events in the department for students, such as Halloween and Christmas parties, and the Spring Formal
- 2018–2022 **Cambridge Hands-On Science**
Volunteered at various outreach events, demonstrating physics experiments to members of the public.

Conferences

2026

- May **Liege Colloquium on Ocean Dynamics: Submesoscale processes in the ocean**
Presentation
- Apr-May **IHP Trimester: Mathematical Developments in Geophysical Fluid Dynamics**
- Apr **CIRM-IHP Joint Program Summer School: Mathematical Developments in Geophysical Fluid Dynamics**
- Feb **Ocean Sciences Meeting**
Presentation

2024

- Feb **Ocean Sciences Meeting**
Poster